

The Doppler-Smith Correspondence: Shared Möbius Structure in Impedance and Frequency Ratios

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Abstract

The classical Smith chart, used as a graphical calculator for complex impedance matching in electrical engineering, and the Doppler frequency shift are both expressible through the same bilinear (Möbius) form, $\Gamma = (u - 1)/(u + 1)$, where u represents normalized impedance $z = Z/Z_0$ or a normalized frequency ratio $1 + \beta$, with $\beta = v/c$. We show this correspondence follows from the general algebraic identity $(A - B)/(A + B) = T(A/B)$ for any two positive quantities A and B , and is therefore a property of the normalized-ratio construction itself, rather than a discovery specific to wave impedance or Doppler kinematics. Recognizing this shared form nonetheless has practical value: standard Smith chart tooling and visualization can be reused directly for Doppler-shift problems, and, as we demonstrate, for acoustic cavity resonance analysis. We derive the correspondence explicitly and verify it against a standard closed-open acoustic tube resonance calculation. We also note where the correspondence's scope ends: it does not, on its own, establish a physical or causal link between impedance matching and relativistic or acoustic Doppler shift.

1 Introduction

1.1 The Smith Chart: A Brief History

The Smith chart, introduced by Philip H. Smith in 1939 [3], is a graphical calculator for solving transmission line problems. It maps normalized impedance $z = r + jx = Z/Z_0$ to the reflection coefficient plane $\Gamma = \Gamma_r + j\Gamma_i$ via the bilinear transform:

$$\Gamma = \frac{z - 1}{z + 1} \tag{1}$$

This mapping is conformal, it preserves angles and maps circles in the impedance plane to circles in the reflection coefficient plane. The chart's grid of constant-resistance circles and constant-reactance arcs enables engineers to determine impedance matches, standing wave ratios, and matching network requirements without solving transmission line equations at every point.

The Smith chart's utility in electrical engineering is well established, as is its mathematical foundation in the bilinear transform and its conformal properties [4, 5]. This paper's contribution is narrower than the chart's engineering utility: we note that the same bilinear form arises naturally when normalizing a Doppler frequency shift, which lets Smith chart visualization tools be repurposed for Doppler and acoustic resonance problems.

1.2 The Taylor-Doppler Identity

The Taylor-Doppler Identity [1] establishes that the classical Doppler formula is the first-order term ($n = 1$) of a Taylor expansion of the phase field along an observer's trajectory on a phase manifold:

$$\Phi(t) = \Phi_0 + \frac{\partial \Phi}{\partial x} \left(vt + \frac{1}{2}at^2 + \frac{1}{6}jt^3 + \dots \right) \quad (2)$$

where v is velocity, a is acceleration, and j is jerk. The present paper does not extend that expansion further. It instead examines the geometric form of the $n = 1$, classical Doppler, term specifically, and shows that its normalized-ratio representation shares the Smith chart's bilinear structure.

1.3 Scope and Contribution

This paper:

- Proves that the Smith chart mapping and a normalized Doppler frequency ratio are both instances of the same Möbius transform $T(u) = (u - 1)/(u + 1)$.
- Shows this correspondence is a consequence of the general identity $(A - B)/(A + B) = T(A/B)$, which holds for any two positive quantities, not a property specific to impedance or Doppler physics.
- Demonstrates that Smith chart visualization extends to acoustic cavity resonance analysis, with a worked example.

This paper does not claim:

- Novelty of the bilinear transform or its algebraic properties.
- That the correspondence reveals a physical or causal connection between impedance matching and Doppler shift, it is a shared algebraic form, applied to two different physical settings.
- Correction of any established Smith chart formalism, or replacement of standard impedance matching or Doppler techniques.

2 Foundational Kinematics and Field Mappings

2.1 Wave Propagation on a Transmission Line

Let a forward-traveling voltage wave propagate through a medium with characteristic impedance Z_0 :

$$V^+(x, t) = V_0 e^{j(\omega t - \beta x)} \quad (3)$$

When this wave encounters a localized impedance mismatch Z_L , a reflected wave is generated:

$$V^-(x, t) = \Gamma V_0 e^{j(\omega t + \beta x)} \quad (4)$$

The reflection coefficient Γ is the ratio of reflected to incident voltage amplitudes:

$$\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0} \quad (5)$$

2.2 The Bilinear Transform

Normalizing impedance by Z_0 , we define $z = Z/Z_0 = r + jx$, yielding:

$$\Gamma = \frac{z - 1}{z + 1} = \frac{(r - 1) + jx}{(r + 1) + jx} \quad (6)$$

2.3 The Doppler Shift Mapping

For a source at rest and an observer moving with normalized velocity $\beta = v/c$, the observed frequency is:

$$f_{\text{obs}} = f_0(1 + \beta) \quad (7)$$

Define $u = f_{\text{obs}}/f_0 = 1 + \beta$, the frequency ratio itself. This is the quantity that plays the role of normalized impedance z in the Doppler setting. Define the normalized Doppler coefficient as:

$$\Gamma_D = \frac{f_{\text{obs}} - f_0}{f_{\text{obs}} + f_0} \quad (8)$$

Substituting (7):

$$\Gamma_D = \frac{f_0(1 + \beta) - f_0}{f_0(1 + \beta) + f_0} = \frac{\beta}{2 + \beta} \quad (9)$$

This is the Doppler mapping, expressed in terms of β . In terms of u , it is simply $\Gamma_D = (u - 1)/(u + 1)$, identical in form to (1). Section 3 makes this identity explicit.

2.4 The Phase Manifold

The 4D phase manifold M is parameterized by:

$$M = \{(x, y, z, t) \in \mathbb{R}^3 \times \mathbb{R}^+\} \quad (10)$$

The scalar phase field $\Phi : M \rightarrow \mathbb{R}$ governs wave propagation. An observer's trajectory is $r(t) = (x(t), y(t), z(t))$. The observed phase is:

$$\Phi_{\text{obs}}(t) = \Phi(r(t), t) \quad (11)$$

The observed frequency is the total time derivative:

$$\omega_{\text{obs}} = \frac{d\Phi_{\text{obs}}}{dt} = \frac{\partial \Phi}{\partial t} + \nabla \Phi \cdot v(t) \quad (12)$$

where $v(t) = dr/dt$ is the observer's velocity.

3 The Doppler-Smith Correspondence

3.1 The General Algebraic Fact

Before stating the correspondence, note the following general identity, which holds for any two positive real quantities A and B :

$$\frac{A - B}{A + B} = \frac{(A/B) - 1}{(A/B) + 1} = T(A/B) \quad (13)$$

This is an algebraic property of the normalized-difference construction itself. It applies to any pair of positive quantities, impedances, frequencies, or otherwise, whenever their difference is normalized by their sum. It carries no physical content on its own.

3.2 Theorem 1 (Shared Möbius Form)

Let Γ be the reflection coefficient for normalized impedance $z = r + jx$, and let Γ_D be the normalized Doppler coefficient for frequency ratio $u = f_{\text{obs}}/f_0 = 1 + \beta$, as defined in (8). Then:

$$\Gamma = T(z), \quad \Gamma_D = T(u) \quad (14)$$

where $T(u) = (u - 1)/(u + 1)$ is the same function in both cases.

Proof. For the Smith chart, $T(z) = (z - 1)/(z + 1)$ by definition (1). For the Doppler coefficient, by (13) with $A = f_{\text{obs}}$, $B = f_0$:

$$\Gamma_D = \frac{f_{\text{obs}} - f_0}{f_{\text{obs}} + f_0} = T(f_{\text{obs}}/f_0) = T(u) \quad (15)$$

Both are evaluations of the same rational function T . Since T is a Möbius transformation, it is conformal on the extended complex plane away from its pole at $u = -1$; both domains considered here, z with $r \geq 0$ and $u = 1 + \beta$ with $|\beta| < 1$, remain bounded away from that point. ■

Remark. This theorem should not be read as revealing a hidden physical connection between wave impedance and Doppler kinematics. It follows directly from (13): any quantity constructed as a normalized difference of two positive values will produce this same form. What Theorem 1 establishes is narrower and more useful in practice: because Γ and Γ_D obey the identical functional form, the graphical and computational apparatus built around the Smith chart, its circles, its unit-disk boundary, its standard construction methods, can be applied directly to Doppler-ratio problems without modification.

3.3 Derivation of Reflection Coefficient Components

Starting from (6), multiply numerator and denominator by the complex conjugate of the denominator:

$$\Gamma = \frac{[(r - 1) + jx][(r + 1) - jx]}{(r + 1)^2 + x^2} \quad (16)$$

Expanding the numerator:

$$\Gamma = \frac{(r^2 - 1) + x^2 + j[x(r + 1) - x(r - 1)]}{(r + 1)^2 + x^2} \quad (17)$$

$$\Gamma = \frac{(r^2 - 1 + x^2) + j(2x)}{(r + 1)^2 + x^2} \quad (18)$$

This yields the explicit real and imaginary components:

$$\Gamma_r = \frac{r^2 - 1 + x^2}{(r + 1)^2 + x^2} \quad (19)$$

$$\Gamma_i = \frac{2x}{(r + 1)^2 + x^2} \quad (20)$$

3.4 The Horizontal Axis Condition

The horizontal axis of the Smith chart is defined by $\Gamma_i = 0$:

$$\Gamma_i = 0 \implies \frac{2x}{(r + 1)^2 + x^2} = 0 \implies x = 0 \quad (21)$$

When $x = 0$, the reflection coefficient simplifies to:

$$\Gamma_r = \frac{r^2 - 1}{(r + 1)^2} = \frac{(r - 1)(r + 1)}{(r + 1)^2} = \frac{r - 1}{r + 1} \quad (22)$$

This is the familiar result: on the horizontal axis, the reflection coefficient is purely real, varying from $\Gamma_r = -1$ (short circuit, $r = 0$) through $\Gamma_r = 0$ (matched load, $r = 1$) to $\Gamma_r = +1$ (open circuit, $r \rightarrow \infty$).

4 Acoustic Application: Cavity Resonance Analysis

4.1 The Acoustic Smith Chart

Theorem 1 lets the Smith chart be used for acoustic cavity analysis. For an acoustic cavity, the relevant impedance is the acoustic impedance $Z_{ac} = p/u_v$, where p is pressure and u_v is particle velocity, using u_v here to avoid collision with the frequency ratio u defined in Section 2.3.

Normalized acoustic impedance:

$$z_{ac} = \frac{Z_{ac}}{Z_0} = r + jx \quad (23)$$

where $Z_0 = \rho c$ is the characteristic acoustic impedance of the medium.

The reflection coefficient at the cavity boundary is:

$$\Gamma_{ac} = \frac{z_{ac} - 1}{z_{ac} + 1} \quad (24)$$

This maps directly onto the Smith chart, with the same geometry as electrical impedance, by Theorem 1.

4.2 Resonance Condition

For an acoustic cavity, resonance occurs when

$$x = 0 \quad (\text{zero reactance}) \quad (25)$$

On the Smith chart, this corresponds to the horizontal axis crossing described in Section 3.4. This is the standard acoustic resonance condition; no additional structural principle is required to explain it.

5 Worked Example: Acoustic Tube Resonance

5.1 Problem Statement

Consider a cylindrical acoustic tube of length $L = 1.0$ m, open at one end and closed at the other, filled with air at room temperature ($c = 343$ m/s). We determine the resonance frequencies using the Smith chart method.

5.2 Traditional Solution

The resonance frequencies for a closed-open tube are:

$$f_n = \frac{(2n-1)c}{4L}, \quad n = 1, 2, 3, \dots \quad (26)$$

For $L = 1.0$ m:

$$f_1 = \frac{343}{4} = 85.75 \text{ Hz} \quad (27)$$

$$f_2 = \frac{3 \times 343}{4} = 257.25 \text{ Hz} \quad (28)$$

$$f_3 = \frac{5 \times 343}{4} = 428.75 \text{ Hz} \quad (29)$$

5.3 Smith Chart Method

The acoustic impedance at the open end of the tube, looking into the tube from outside, is:

$$Z_{\text{in}} = -jZ_0 \cot(kL) \quad (30)$$

where $k = 2\pi f/c$ is the wavenumber. Normalized impedance:

$$z_{\text{in}} = -j \cot(kL) \quad (31)$$

At resonance, $x = 0$, which occurs when $\cot(kL) = 0$:

$$kL = \frac{(2n-1)\pi}{2} \quad (32)$$

$$\frac{2\pi fL}{c} = \frac{(2n-1)\pi}{2} \quad (33)$$

$$f_n = \frac{(2n-1)c}{4L} \quad (34)$$

This matches the traditional result. The Smith chart plot of z_{in} shows the trajectory crossing the horizontal axis at precisely these frequencies.

5.4 Smith Chart Trajectory

Frequency	kL	z_{in}	Γ	Position on Chart
$f_1 = 85.75 \text{ Hz}$	$\pi/2$	0	-1	Short circuit $(-1, 0)$
$f_2 = 257.25 \text{ Hz}$	$3\pi/2$	0	-1	Short circuit $(-1, 0)$
$f_3 = 428.75 \text{ Hz}$	$5\pi/2$	0	-1	Short circuit $(-1, 0)$

Interpretation: Each resonance is a horizontal-axis crossing, corresponding to the standard quarter-wavelength boundary condition $kL = (2n-1)\pi/2$ for a closed-open tube. At resonance, the input impedance seen at the open end approaches the Smith chart's short-circuit point, consistent with the standing-wave boundary condition at the closed end. This is a direct consequence of the boundary conditions at the open and closed ends of the tube.

6 Discussion

6.1 What the Correspondence Shows

Γ and Γ_D share a functional form because both are constructed as normalized differences of two positive quantities. Recognizing this lets Smith chart tooling, built and refined over eight decades of electrical engineering practice, be reused for Doppler and acoustic problems without rebuilding equivalent tooling from scratch. That is the paper's contribution: practical and interpretive, not a claim of new physics.

6.2 Implications for Acoustic Engineering

Acoustic engineers can use the Smith chart for cavity resonance analysis, impedance matching layer design, standing wave visualization, and multi-layer cascade analysis, exactly as in the electrical case, since Section 4 establishes the mapping is identical.

7 Conclusion

This paper has established that the Smith chart's reflection coefficient and a normalized Doppler frequency ratio are both instances of the same Möbius transform, a consequence of the general algebraic identity $(A-B)/(A+B) = T(A/B)$. This shared form is not evidence of a physical connection between impedance and Doppler shift; it is a property of the normalization itself, applied

here to two different physical quantities. The correspondence is algebraic, not physical, and that is exactly what makes it useful: it lets one set of tools serve two domains. The worked example in Section 5 confirms the mapping holds exactly against the standard closed-open acoustic tube result.

The value of the correspondence is therefore not that it unifies different physical phenomena, but that it identifies a shared mathematical language. Once that language is recognized, the same graphical and computational tools can be applied wherever the underlying structure is the same, while their physical interpretation remains entirely domain-specific.

References

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